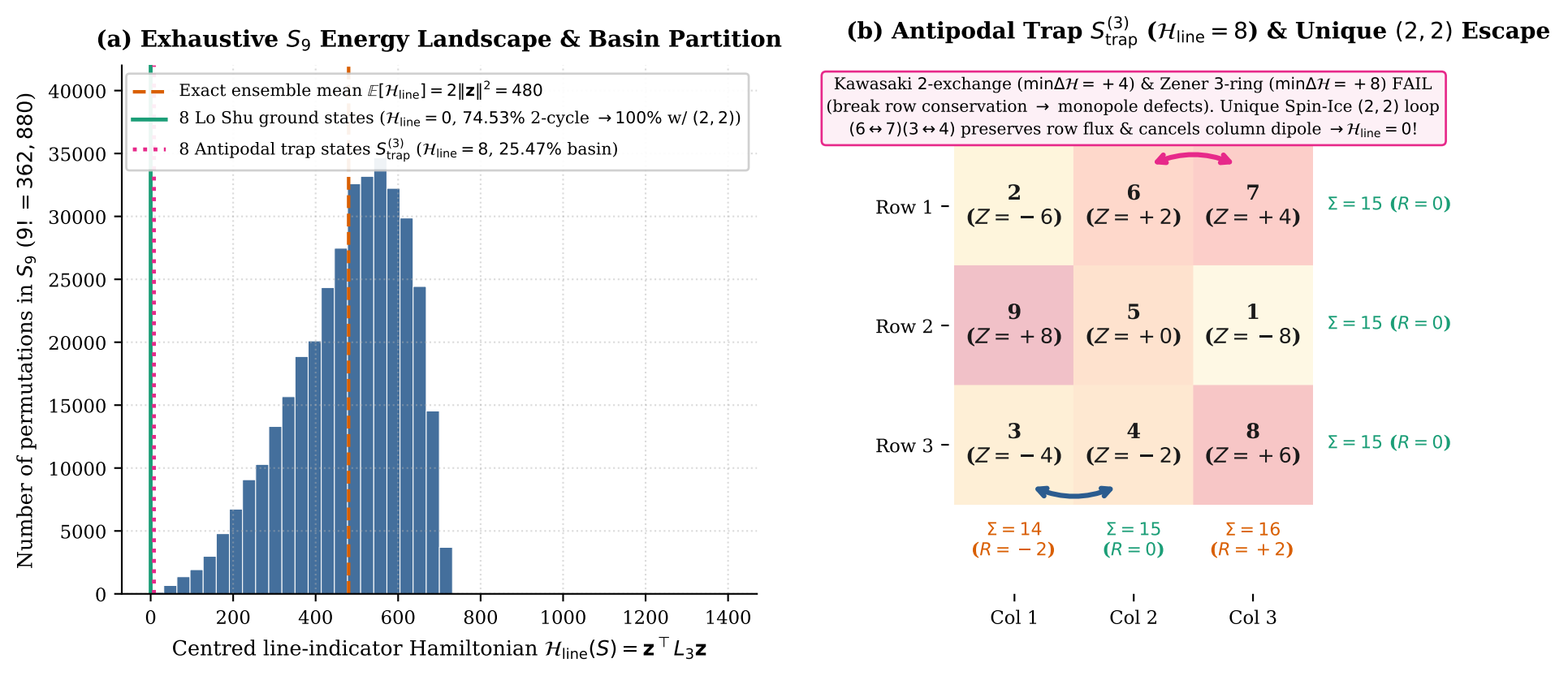

(a) Exhaustive $S_9$ Energy Landscape & Basin Partition
Exact ensemble mean $\mathbb{E}[\mathcal{H}_{\text{line}}] = 2\|\mathbf{z}\|^2 = 480$
8 Lo Shu ground states ($\mathcal{H}_{\text{line}} = 0$, 74.53% 2-cycle $\rightarrow$100% w/ (2, 2))
8 Antipodal trap states $S_{\text{trap}}^{(3)}$ ($\mathcal{H}_{\text{line}} = 8$, 25.47% basin)
Number of permutations in $S_9$ (9! = 362, 880)
0
5000
10000
15000
20000
25000
30000
35000
40000
0
200
400
600
800
1000
1200
1400
Centred line-indicator Hamiltonian $\mathcal{H}_{\text{line}}(S) = \mathbf{z}^\top L_3 \mathbf{z}$
(b) Antipodal Trap $S_{\text{trap}}^{(3)}$ ($\mathcal{H}_{\text{line}} = 8$) & Unique (2, 2) Escape
Kawasaki 2-exchange (min$\Delta\mathcal{H} = +4$) & Zener 3-ring (min$\Delta\mathcal{H} = +8$) FAIL
(break row conservation $\rightarrow$ monopole defects). Unique Spin-Ice (2, 2) loop
$(6 \leftrightarrow 7)(3 \leftrightarrow 4)$ preserves row flux & cancels column dipole $\rightarrow \mathcal{H}_{\text{line}} = 0$!
Row 1
Row 2
Row 3
2 ($Z = -6$)
6 ($Z = +2$)
7 ($Z = +4$)
9 ($Z = +8$)
5 ($Z = +0$)
1 ($Z = -8$)
3 ($Z = -4$)
4 ($Z = -2$)
8 ($Z = +6$)
$\Sigma = 15$ ($R = 0$)
$\Sigma = 15$ ($R = 0$)
$\Sigma = 15$ ($R = 0$)
$\Sigma = 14$ ($R = -2$)
$\Sigma = 15$ ($R = 0$)
$\Sigma = 16$ ($R = +2$)
Col 1
Col 2
Col 3